

PLUMBING

1. Scope:

A. Work included: All labor, materials, appliances, tools, equipment, facilities, transportation and services necessary for and incidental to performing all operations in connection with furnishing, delivery and installation of the work of this section, complete, as shown on the drawings and/or specified herein. Work includes, but is not necessarily limited to, the following:

1. Examine all other sections for work related to those other sections and required to be included as work under this section.
2. Sanitary soil, waste and vent and condensate drain systems.
3. Domestic hot and cold water systems.
4. Hot water heater, and recirculating system and flues as required.
5. Fuel gas systems.
6. All plumbing fixtures, trim, valves, traps, drains, clean outs, access plates or boxes, hose bibbs, floor sinks, floor drains, and other miscellaneous items required for complete installation.
7. Connections to sanitary sewer main, and connections to water and gas meters as indicated on the drawings.
8. Excavation and backfilling.
9. Roughin and connection to equipment furnished by others.
10. All cutting and patching in stud walls required for installation.
11. Coordination of holes in concrete.

B. Related work not included in this section:

1. All electric wiring and controls.
2. Concrete work.
3. Painting.
4. Landscape irrigation.
5. Concrete coring.

2. Ordinances and Regulations:

A. All apparatus and materials installed under this contract shall be installed to comply with any and all requirements of local ordinances, and all other authorities having jurisdiction.

3. Permits, Licenses, Patent Rights and Inspections:

A. Shall be obtained and paid for by this contractor. He shall furnish signed, certified certified and acceptable copies to the Architect for his records. The Owner will prepare, submit and pay plan check drawings required.

4. Locations:

- A. No work shall be exposed unless so indicated on drawings or as otherwise approved in writing by the Architect.
- B. Avoid interference with Architectural and Structural features and work of other trades, preserving headroom and keeping openings and passageways clear.

5. List of Materials, Fixtures and Equipment:

- A. The Contractor shall submit to the Architect for his review, within fifteen (15) brochures, each giving a complete list of materials, fixtures and equipment with names and addresses of manufacturers, catalog numbers and trade names.
- B. Any items noted by the Architect as not conforming to the specifications or not of proper quality or grade shall be deleted and suitable substitution in lieu thereof, shall be provided as designated by the Architect.

6. Protection and Safeguards:

- A. Erect and maintain suitable barriers, protective devices, lights and warning signs where required for the protection of the public and workmen around the building.
- B. Plug and cap all piping during installation and keep free from debris and other contamination.

7. Materials and Workmanship:

- A. All materials shall be new, and in perfect condition. Materials for similar uses shall be of same type and manufacture.
- B. Workmanship shall be best of standard practice of the trade.
- C. Trim, fittings and pipe in finished spaces shall be chromium plated brass or covered with chromium plated brass sleeves.

8. Utilities Connections:

A. This Contractor shall install sewer, water, and gas system complete, and shall make final connections to sewer main in street, and connections to gas and water meters furnished by others; all as indicated on the drawings. Plumbing Contractor shall pay all utility, meter, and connection charges.

9. Openings:

- A. Openings are prohibited in any structural element unless indicated on drawings or without written authorization from the Architect.
- B. The Contractor shall cooperate with other trades in providing information as to openings required in walls, floors and foundations for pipe lines and equipment.
- C. Sleeves shall be accurately located and placed in forms before concrete is poured.
- D. Where several pipes are located in close proximity, the contractor may provide one opening for all pipes in lieu of several openings. All openings not indicated on the drawings must be approved by the Architect.
- E. The Contractor shall pay all extra costs for cutting of holes as a result of incorrect, delayed or neglected location of sleeves or openings.

10. Waterproofing:

A. Make absolutely water tight any openings through waterproofed construction caused by the penetration of plumbing work in a manner approved by the Architect.

11. Excavation and Back fill:

- A. This Contractor shall do all excavation necessary for installation of plumbing system.
- B. Run trenches in straight lines with bottom properly pitched to give support to piping along its full length.
- C. Upon completion of work, all trenches shall be back filled evenly, and thoroughly compacted to 95% density using acceptable fill materials.

12. Pipe Schedule:

A. Soil, waste & storm drain lines:

1. Service weight cast iron soil pipe and fittings, as approved by local code.
2. Underground soil and waste line: Service weight cast iron soil pipe and fittings, ABS plastic pipe and fittings as approved by local code.
3. Yard piping: From points 5 feet outside of building to street connection shall be extra heavy vitrified clay pipe and fittings.

B. Vent piping:

1. 1 1/2" and smaller: Galvanized steel pipe, schedule 40, with standard black cast iron screwed fittings to 6" above floor only.
2. DWV copper same as specified under soil and waste, as approved by local code.
3. Underground vents: Service weight cast iron soil pipe and fittings same as specified under soil and waste, as approved by local code.

C. Water and condensate drain piping:

1. Above ground: Type "M" copper with sweat soldered fittings.
2. Below grade: The "L" copper, hard, with sweat soldered fittings, sized 3" and smaller.
3. On site mains, 4" and larger, asbestos cement pressure pipe with cast iron ring tight fittings as approved by local code.
4. Condensate drain piping: Type "M" copper with 3" minimum hydraulic trap seal and shall be same size equipment connection unless otherwise noted.

D. Fuel gas piping:

1. Black steel pipe, schedule 40, with standard weight malleable iron 150# screwed fittings.

13. Pipe Installation:

A. General:

1. All piping shall run parallel to building construction, shall be neat and workmanlike.
2. Unless otherwise specified, all pipes shall be concealed in walls, partitions, above ceiling or furled spaces.
3. All piping shall be properly supported and pitched to drain.
4. All piping shall have minimum clearance of 1/2" and shall be isolated from building structure with no less than 1/2" thick waffle pattern hair felt, treated to prevent mildew, or approved suspension clamps.
5. Where piping is at the exterior of the building, it shall be a minimum of 12" below finish grade. No pipes shall be exposed on the exterior of building.
6. All gas pipes and fittings below grade shall be protected with Hunt's Process #67, or Scotch coated.
7. All pipes passing through roof shall be flashed with 26 gauge galvanized iron assembly set in waterproof mastic compound. All flashing shall be caulked around pipe.
8. Connection between copper or brass piping and ferrous material shall be made with approved dielectric couplings.
9. Avoid connections and equipment installation that could cause a cross connection between water supply and drainage resulting in contamination of said water pipe.

B. Soil, Waste, Vent Storm Drainage and Condensate Drain Piping:

1. Run level without pockets and as straight as possible.
2. Bends are not permitted, offsets made with fittings.
3. Hot and Cold water lines at least 6" apart where piping is parallel.
4. Expansion and contraction provided for by offsets and loops or swing joints.
5. Make all connections to gas burning equipment and appliances with gas cock and flexible connector.

C. Cast Iron Soil, Pipe Joints:

1. No hub piping shall be connected with approved stainless steel shields and clamps with neoprene gasket and shall comply with the cast iron pipe institute standards 301895.

D. Screwed Pipe Joints:

1. Cut threads with new, clean dies, full thickness and carefully ream.
2. Apply mineral oil smoothly to male threads and to threads left exposed after fabrication.

E. Copper or Brass Joints:

1. Copper or brass solder fittings shall be thoroughly cleaned with "Scotchbrite" and assembled with 5050 hard solder and noncorrosive flux.

F. Pipe excutcheon plates:

1. All exposed water, waste and gas piping passing through floors, walls and ceilings shall have excutcheon plates. Plates shall be beaten & Cad well #3A cast brass, chrome plated, split hinged.

14. Valves and specialties:

A. General:

1. Provide valves in the water and gas systems, where required by code, so located, arranged and operated as to give complete regulation of apparatus, equipment and fixtures; valves to accomplish this shall be provided whether shown on this drawings or not. Valves for similar service shall be of one manufacturer. Valves shall be Nibco, Stockham, Homestead, Crane, Jenkins or Price Pfister.

15. Hangers and Supports:

A. Shall be furnished and installed as required by code. All water pipe shall be sound isolated from building with 1/4" hard felt pads treated against mildew. Support and fasten copper water pipes as follows:

1/2" Diameter	5 ft. maximum
3/4" and 1" diameter	6 ft. maximum
1 1/4" thru 2" diameter	8 ft. maximum
2 1/2" thru 4" diameter	10 ft. maximum

B. Use approved pipe clamps resting on 1/2" neoprene at base of all risers.

C. Concrete inserts: Fee and mason, #185 insert.

16. Access Plates:

A. Provide and install access plates where necessary and where required by code, including those indicated on the drawings. The type and finish shall be approved by the Architect.

17. Clean outs:

A. Clean outs shall be installed as required by code. Interior floor clean outs shall be provided with non slip cover plates.

18. Drains:

A. Floor drains: J.R. Smith, Josam or Zurn, with cast iron bodies, clamping collar, and grate as follows:

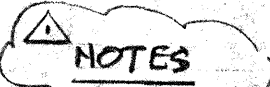
19. Fittings:

- A. Malleable iron screwed fittings: standard weight, flat bank type, cane, grinnell or walworth.
- B. Cast iron drainage fittings: for iron pipe sizes screwed connections, specially recessed type, crane, grinnell or cal American, hot asphalt coated.
- C. Unions: malleable iron screwed, brass to iron set (steel pipe), copper to copper or brass seat (copper pipe).
- D. Hose Bibbs Item HB1: 3/4" rough brass hose bibb, pricepfister, acorn or J.R. Smith, with vacuum breaker.
- E. Trap primers: E & S or Chicago, as required.

20. Plumbing Fixtures and Equipment:

A. General:

1. All fixtures shall be installed complete ready for use, with all required faucets, strainers, pop ups and traps.
2. Provide all fixtures and equipment with stop valves on water service. Chrome plate all exposed metal items in toilet rooms.



1. ALL MATERIALS USED IN THE NEW WATER SUPPLY SYSTEM EXCEPT VALVES AND SIMILAR DEVICES SHALL BE OF A LIKE MATERIAL USED IN THE EXISTING BUILDING PIPING.

2. MIN. DRAIN SLOPE 2%

3. MAX GPF OF TOILET 1.6

4. PROVIDE CO MAX EVERY 100'

5. SEE NOTE 21 SH. M.O.I FOR OPERATING - SERVICE - MAINTENANCE BROCHURE SUBMITTAL REQUIREMENTS, TYP FOR PLUMBING EQUIPMENT.

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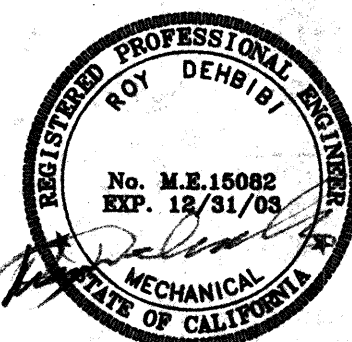
FOR

SAINT EDMUND'S

EPISCOPAL CHURCH

1175 SAN GABRIEL BLVD., SAN MARINO, CA., 91108-8048

STAMPS



REVISIONS

DESCRIPTION	DATE BY
PLAN CHECK	1/4/0

PROJECT NUMBER

2000

SCALE

AS NOTED

PRINTED

11/12/00

TITLE

PLUMBING
SPECIFICATION
SHEET

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35 OF 44